

Experiment 10

A* Algorithm - Python Program

Aim

To write a Python program to implement the A* algorithm to find the shortest path between two nodes.

Algorithm

1. Start.
2. Initialize the graph and heuristic values.
3. Set the start and goal nodes.
4. Add the start node to the open list.
5. Select the node with the lowest $f = g + h$.
6. Expand its neighboring nodes and update their costs.
7. Repeat until the goal node is reached.
8. Display the shortest path.
9. Stop.

Program

```
graph = {  
    'A': [('B', 1), ('C', 3)],  
    'B': [('D', 3), ('E', 6)],  
    'C': [('F', 5)],  
    'D': [('G', 2)],  
    'E': [('G', 2)],  
    'F': [('G', 1)],  
    'G': []  
}
```

```
h = {'A':7, 'B':6, 'C':4, 'D':2, 'E':1, 'F':1, 'G':0}
```

```
open = [(h['A'], 0, 'A', ['A'])]
```

```
while open:
```

```
    open.sort()
```

```
    f, g, node, path = open.pop(0)
```

```
    if node == 'G':
```

```
        print("Shortest Path:", path)
```

```
        print("Cost:", g)
```

```
        break
```

```
    for nxt, cost in graph[node]:
```

```
        open.append((g + cost + h[nxt], g + cost, nxt, path + [nxt]))
```

Input

(No input required)

Output

```
----- RESULT: SUCCESS. -----  
Shortest Path: ['A', 'B', 'D', 'G']  
Cost: 6
```

Result

Thus, the Python program to implement the A* algorithm was executed successfully, and the shortest path from the start node to the goal node was obtained.